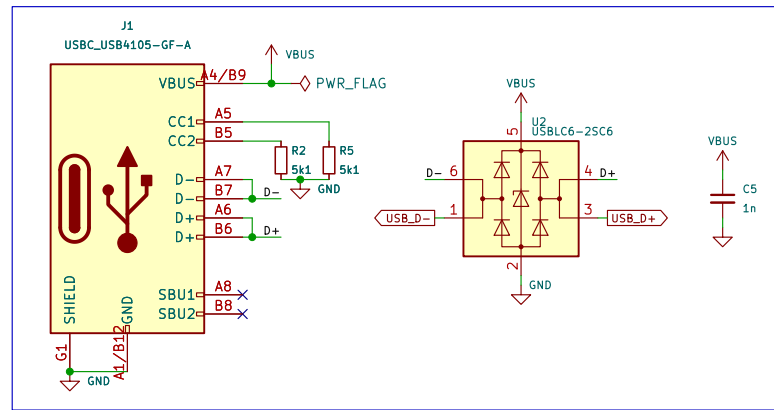
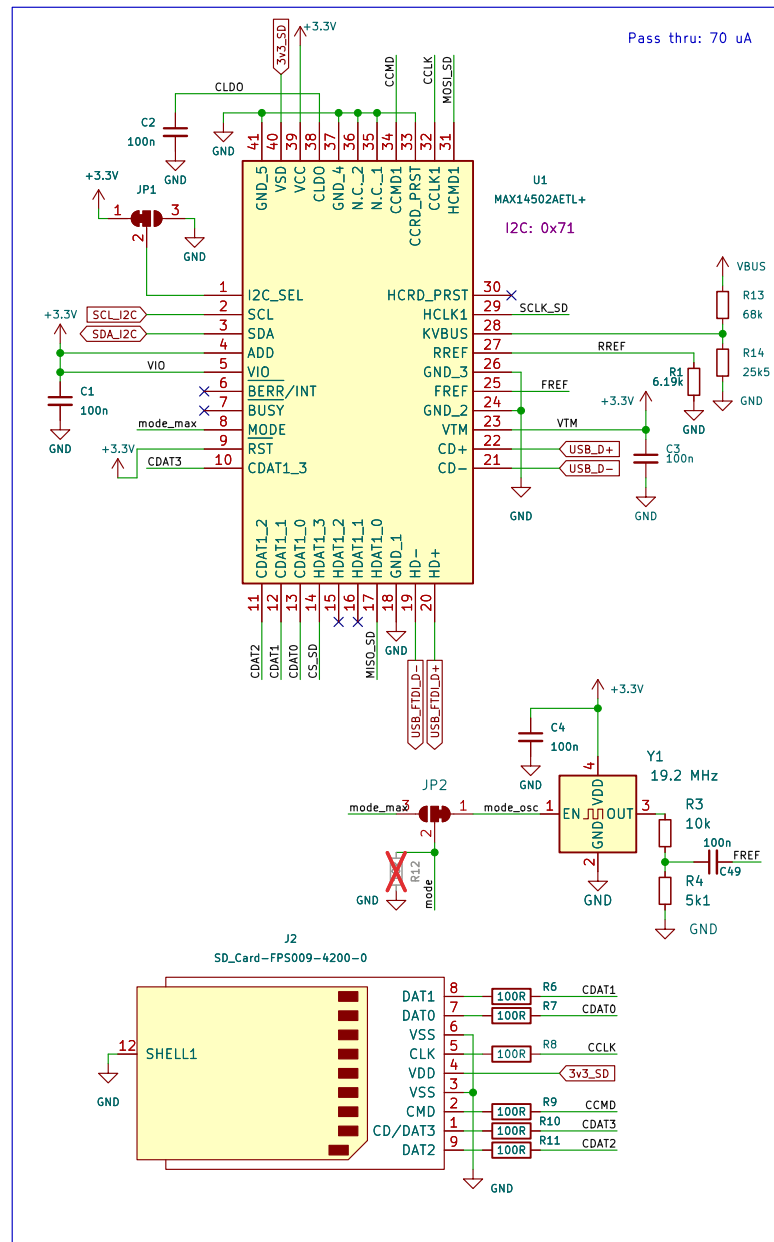


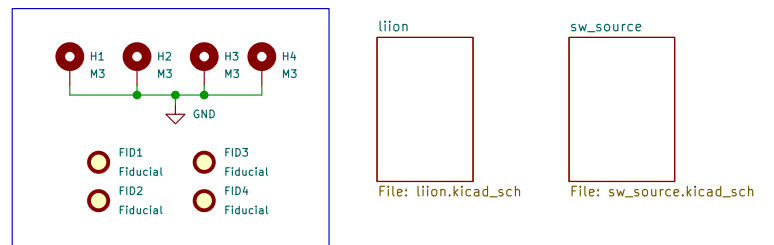
USB-C connector and ESD protection



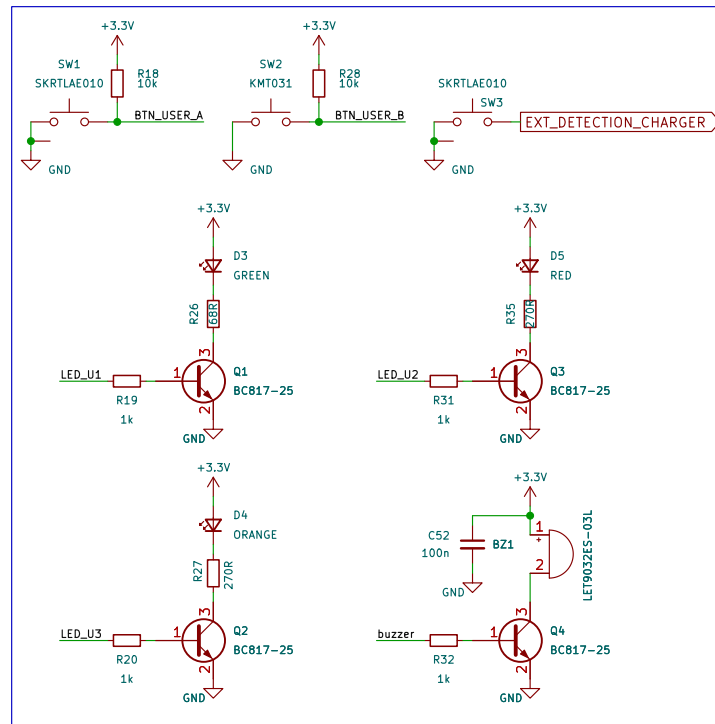
SD card interface (USB to SDCard reader and USB-bridge to FTDI)



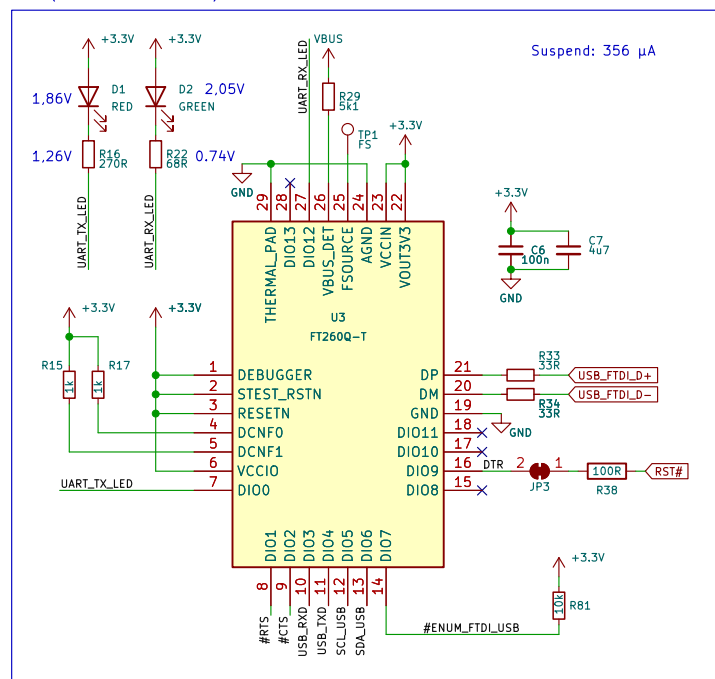
Mechanical



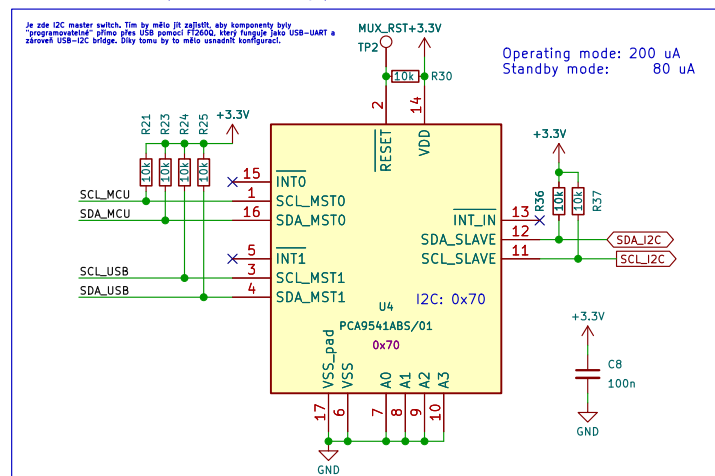
User interface (LED indicators, Buttons)



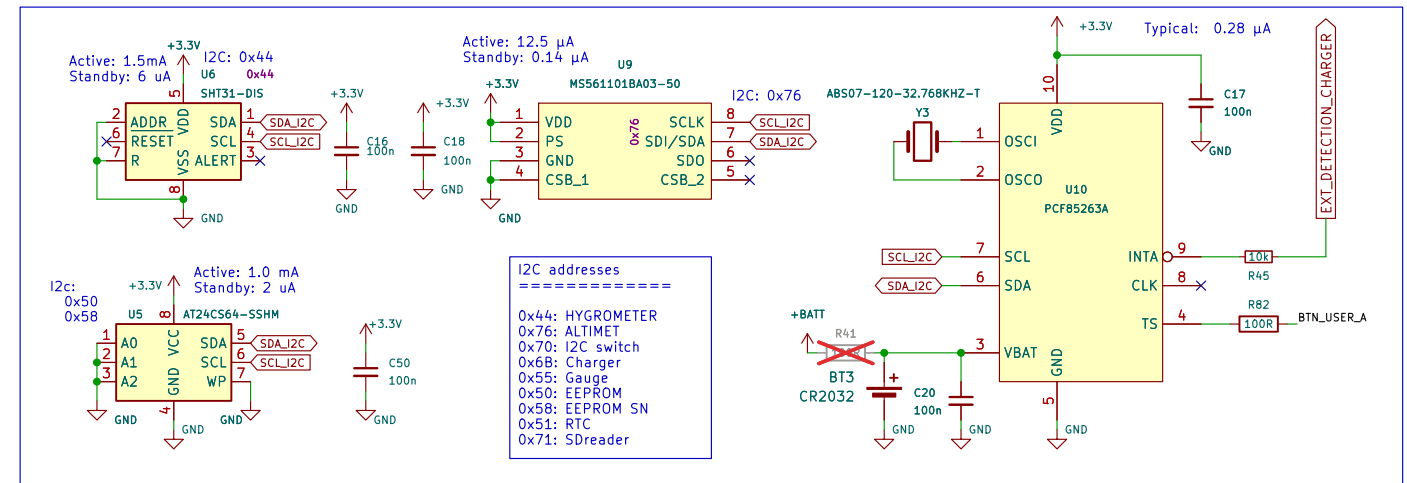
FTDI (USB to UART and I2C)



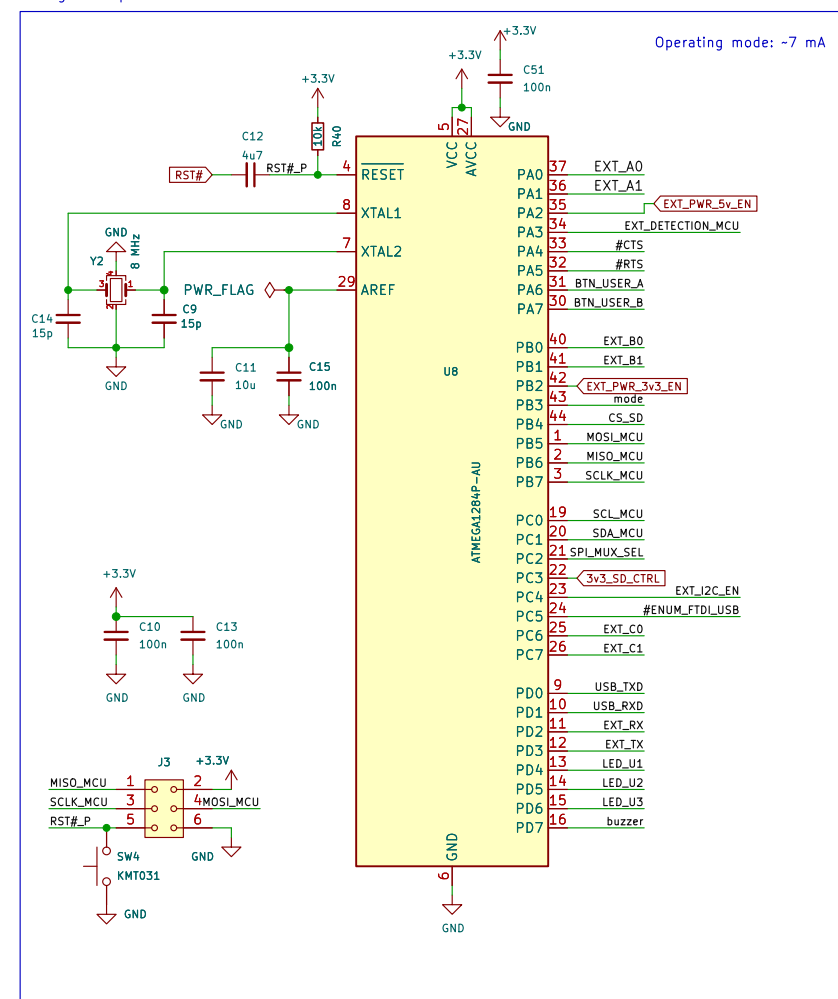
I2C master selector (from FTDI or ATmega)



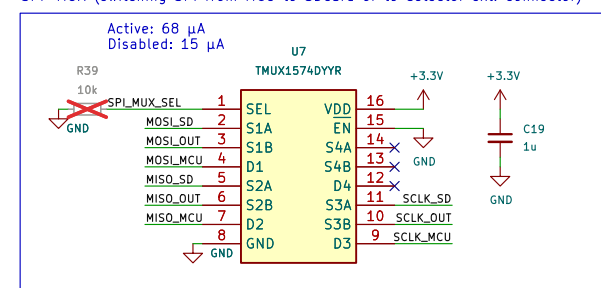
I2C sensors and accessories (hygrometer, altimeter, eeprom, RTC)



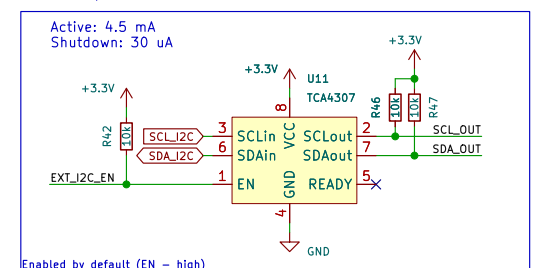
ATmega 1284p microcontroller



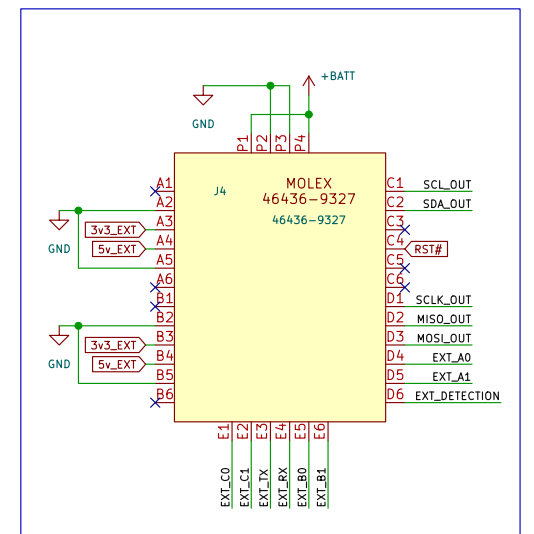
SPI-MUX (switching SPI from MCU to SDcard or to detector ext. connector)



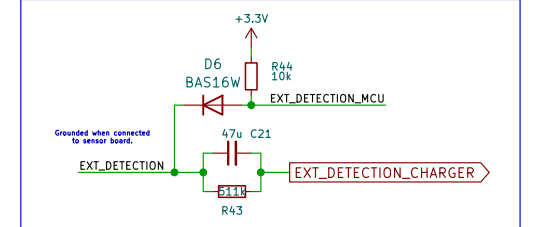
I2C buffer, allows to unconnect external I2C



Connector to sensor board



Auto-power up circuit



Title: BATDATUNIT01D

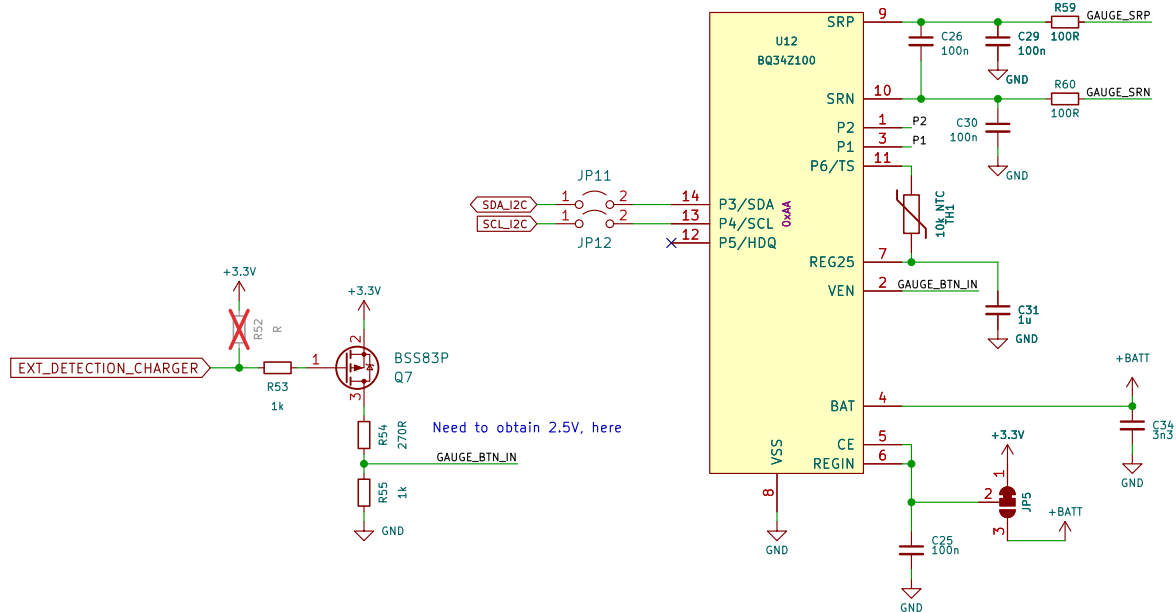
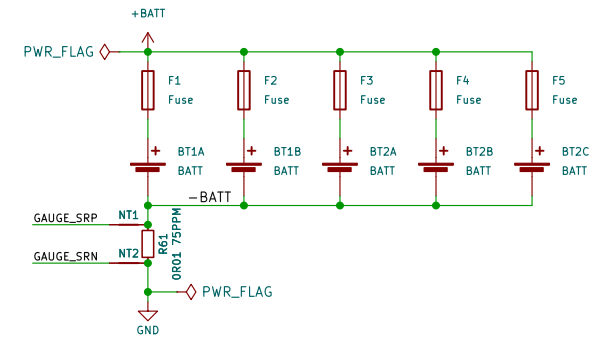
Universal power and data storage unit.

Author: MLAB.cz, roman-dvorak

Size: A3 Page: 1/3 Date: 2026-09-27

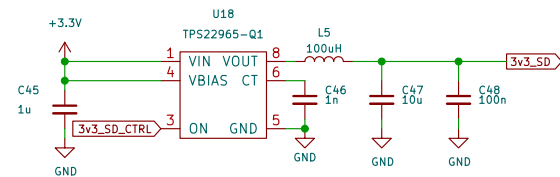
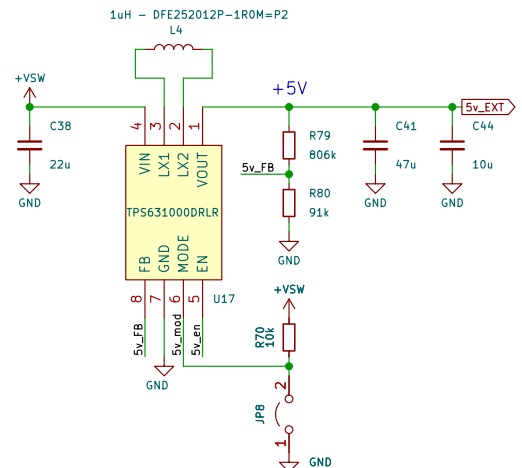
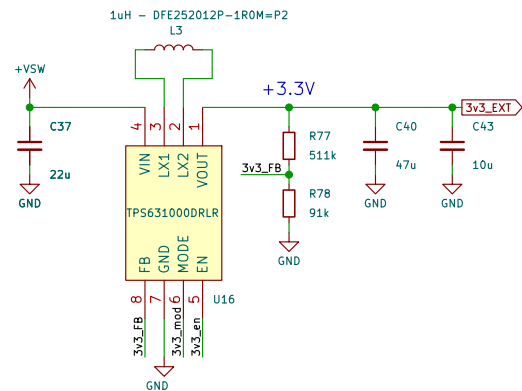
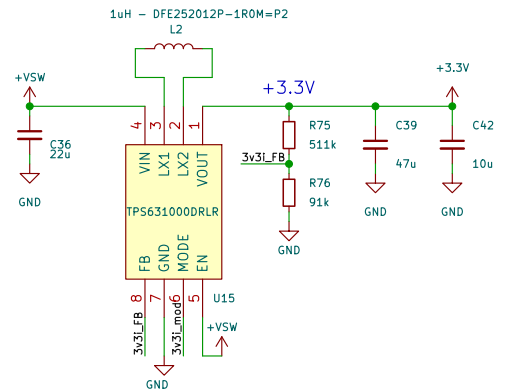


FUEL GAUGE



Date: 2026-09-27





3 switching sources

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3v3 – Internal

For powering internal electronics

3v3 – External

For powering external devices with +3V3

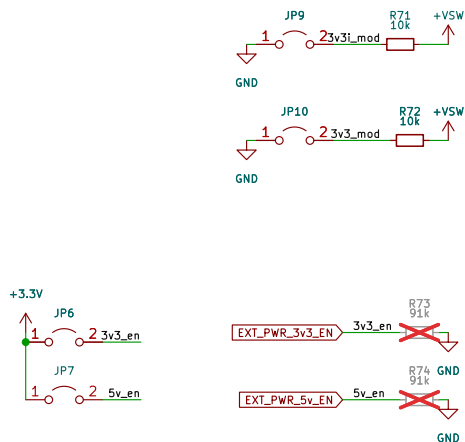
5v – External

For powering external devices with +5V

Internal 3v3 is enabled after physical power up.

External (3v3 and 5v) is by default powered off. Can be switched on by CPU or permanently by solder-jumper.

All switching converters are by default in forced PWM mode (not in power save mode). It results in cleaner voltage output.



Title: BATDATUNIT01D

Universal battery and data storage unit.

Author: MLAB.cz, roman-dvorak

Size: A4

Page: 4/3

Date: 2026-09-27

